Review retrospective:
the quartic–sextic moment specialisation — two dead opens, a no-op, and a mislabelled proof step

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Abstract

A soundness review of `Laplace/TwoD/QuarticSexticMomentAsymptotic.lean`: the $(k_1, k_2) = (2, 3)$ specialisation of the 2D separable mixed-even Gibbs moment $\langle x^{2j} y^{2k} \rangle_{2D}$. Result: **a clean fidelity pass with four edits** — dead `open Real` *and* `open MeasureTheory`, a literal no-op rewrite, and a docstring that named the wrong proof step. The richest cleanup of the `QuarticSextic*` batch.

1 Setting

The moment counterpart of the partition and numerator specialisations, and the $(2, 3)$ instance of the generic moment lift. It repackages the factorisation theorem (T1, `gibbsExpectation_quarticSextic_pow_pow_eq`) as the closed form, rescaled limit, and equivalence.

2 What was reviewed

Three public theorems plus the constant $C(j, k) = 24^{j/2}\, 720^{k/3} \frac{\Gamma((2j+1)/4)}{\Gamma(1/4)} \frac{\Gamma((2k+1)/6)}{\Gamma(1/6)}$, the $(2, 3)$ instance of the generic `kthKthMomentConst`, exponent $-(j/2 + k/3)$.

3 Fidelity / soundness findings

Sound; no theorem-level mismatch. Constant and exponent correct, `ht : 0 < t` present, eventual positivity used for the `atTop` statements. **One docstring inaccuracy, fixed.** The `isEquivalent_rpow` docstring claimed the proof goes "via `Asymptotics.IsEquivalent.refl` after `tendsto_congr'`", but the proof actually uses `IsEquivalent.refl.congr_left` after a `filter_upwards` eventual-equality — a wrong lemma named in proof-strategy prose. Corrected to match the proof (comment-only). This is the same "prose that names a mechanism is a checkable claim" lesson as the generic monomial "used twice" miscount.

4 Simplifications applied

Three. Both `open Real` (all `Real.*` qualified) *and* `open MeasureTheory` are dead — like the partition specialisation, this file states the moment via the `gibbsExpectation` *def*, with no bare

$\int$, so the measure-theory notation is never used. And a literal no-op `rw [show -(j/2+k/3) = -(j/2+k/3) from rfl]` (an $X \to X$ rewrite) was removed. Module green at 2703 jobs, no warnings, signatures byte-identical. GPT affirmed.

5 Disagreements and open questions

None.

6 Lessons

- **The def-vs-bare-$\int$ distinction predicts the second dead open precisely.** The partition and moment specialisations (`partitionFunction`/`gibbsExpectation` defs, no bare $\int$) shed *both* opens; the numerator (bare $\int$) keeps `MeasureTheory`. A one-line `grep` for bare $\int$ on a `TwoD/` file calls the dead-open count before the build does.
- **Proof-strategy prose drifts from the proof.** A docstring naming `tendsto_congr'` for a proof that uses `congr_left` is a small but real fidelity miss — the kind that only an actual read of the proof body catches, and worth fixing because it misleads the next reader about which API does the work.

7 Follow-ups

- The `QuarticSextic*` closed-form cores remain: `QuarticSextic` (the seabed closed form the asymptotic files consume), `QuarticSexticMoments`, `QuarticSexticOddMoments`.
- Batched hygiene tide: the `t>0` module-docstring clauses across `TwoD/`, plus the `show`→`change` proof-golf the reviewer keeps flagging.